Lecture 8.

Principle of Resolution.

Literal: A propositional variable or its Negation.

s, $\neg r$.

Clause: A disjunction of literals.

$\neg t \vee q$. $\quad \neg s \vee q \vee \neg r \vee p \vee z$.

$P_1 \longrightarrow C_1$

$P_2 \longrightarrow C_2$

$\vdots \qquad \vdots$

$P_N \longrightarrow C_M$. $M \geq N$.

$\therefore C \longrightarrow C_{M+1}$.

Taking Negation.

Combine Clauses Repeatedly Using Resolution Unless you obtain empty clause.

$\neg, \wedge, \vee, \rightarrow, \leftrightarrow$.

$p \wedge q$ — p — (4); q — (5).

$p \rightarrow q = \neg p \vee q$. — (3)

$p \leftrightarrow q = p \rightarrow q \wedge q \rightarrow p$.

$p \rightarrow q = \neg p \vee q$. — (1)

$q \rightarrow p = \neg q \vee p$. — (2)

$p \vee q = p \vee q$.

Ex.

p — P1
$p \to q$ — P2
$\therefore q$ — C.

p — C1 ✓
$\neg p \vee q$ — C2 ✓
$\neg q$ — C3 ✓

$p \vee q$
$\neg p \vee r$
$\therefore q \vee r$

Resolution.

from C1, C2 q — (4) ✓
from C3, 4 $\square$ — (5)

Ex 11
65

$T \to M \vee E$ — P1
$S \to \neg E$ — P2
$T \wedge S$ — P3
$\therefore M$

$\neg T \vee M \vee E$ — C1 ✓
$\neg S \vee \neg E$ — C2 ✓
T — C3 ✓
S — C4 ✓
$\neg M$ — C5 ✓

from C1, C2 $\neg T \vee M \vee \neg S$ — (6) ✓
from C3, 6 $M \vee \neg S$ — (7) ✓
" C4, 7 M — (8) ✓
" C5, 8 $\square$ — (9)

Therefore Argument is Valid.

Ex 6 :-
PG2

Ex6 :-
PG2

$\neg p \wedge q$ — P1

$r \to p$ — P2.

$\neg r \to s$ — P3.

$s \to t$ — P4

$\therefore t$.

$\neg p$ — C1 ✓

q — C2

$\neg r \vee p$ — C3. ✓

$r \vee s$ — C4 ✓

$\neg s \vee t$ — C5 ✓

$\neg t$ — C6 ✓

from C4 C3 $\neg r$ — 7 ✓

" C4 7 s — 8 ✓

" C5, 8 t — 9 ✓

" C6, 9 $\square$ — 10

therefore it is Valid.

E7
P6

$p \to q$ — P1

$\neg p \to r$ — P2

$r \to s$ — P3

$\therefore \neg q \to s$. — C.

$\neg p \vee q$ — C1. ✓

$p \vee r$ — C2. ✓

$\neg r \vee s$ — C3. ✓

$\neg q$ — C4 ✓

$\neg s$ — C5 ✓

$\neg(\neg q \to s)$.

$\neg(q \vee s)$.

$\neg q \wedge \neg s$. — $\neg q$

— $\neg s$.

from C1, C2 $q \vee r$ — 6 ✓

" C3, 6 $q \vee s$ — 7 ✓

" C4, 7 s — 8. ✓

" C5, 8 $\square$ — 9

Valid.

Quiz 2 :-

19-Sep-2025.

$\forall x\, p(x)$ $\quad\quad$ $x \in \{1, 0\}$.

$\exists x\, \neg p(x)$.

Proof if valid.

Note:- $p(1)$ is a proposition.
$p(2)$ " " proposition.

Solution:-

$\forall x\, p(x)$ — p_1 $\quad\quad$ $x \in \{1, 0\}$.

$\exists x\, \neg p(x)$ — p_2.

$p(1) \wedge p(0)$ → $p(1)$ — c_1

$p(0)$ — c_2

$\neg p(1) \vee \neg p(0)$. → $\neg p(1) \vee \neg p(0)$ — c_3.

from (c_1, c_3) $\neg p(0)$. — (4)

from $(c_2, 4)$ $\square$. Valid.